

Exact Maximum Loneliness in the Family $(1, 4, 5, 6, 7, m)$

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

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Abstract

For a positive integer $m \notin \{1, 4, 5, 6, 7\}$, let

$$L(m) = \max_{t \in \mathbb{R}/\mathbb{Z}} \min_{v \in \{1, 4, 5, 6, 7, m\}} \|tv\|.$$

Write $f_m(t)$ for the inner minimum. We determine $L(m)$ and every maximizing time for all admissible m . If $11 \nmid m$, then $L(m) = 2/11$, with the maximizers determined by $m \bmod 11$. At $m = 11$ and $m = 22$ the value is $2/13$. If $m = 11n$ with $n \geq 3$, then

$$L(m) = \frac{2n}{11n+4}, \quad \{t : f_m(t) = L(m)\} = \left\{ \frac{5n+2}{11n+4}, \frac{6n+2}{11n+4} \right\}.$$

Consequently, under the strict convention $1/7 < L(m) < 1/6$, the near-tight parameters are exactly $m = 11, 22, 33$. The proof is an exact decomposition of a superlevel set of the base quintuple, followed by a uniform endpoint argument; no finite cutoff is used.

1 Introduction

For $x \in \mathbb{R}$, write

$$\|x\| = \min_{k \in \mathbb{Z}} |x - k|,$$

and identify a time with its class in the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For a finite set V of distinct positive integer speeds, its maximum loneliness is

$$\text{ML}(V) = \max_{t \in \mathbb{T}} \min_{v \in V} \|tv\|.$$

The maximum exists by continuity and compactness.

This note concerns the one-parameter family $V_m = \{1, 4, 5, 6, 7, m\}$. Cordella's study of the six-speed lonely-runner spectrum gives a finite-exception structural theorem for near-tight sextuples, an exact collision/antipode candidate-time criterion, and an exhaustive search through maximum speed 110 [1, Lemma 2.2 and Theorem 6.1]. Its Table 2 records

$$\text{ML}(V_{22}) = \frac{2}{13}, \quad \text{ML}(V_{33}) = \frac{6}{37},$$

and the accompanying discussion notes that the base quintuple $\{1, 4, 5, 6, 7\}$ has maximum loneliness $2/11$ [1, Table 2 and §6.1]. The source does not give an all-parameter formula for V_m . The result below supplies that formula and the complete equality classification. This comparison is limited to the cited source; no broader priority claim is made.

Put

$$D = \mathbb{Z}_{>0} \setminus \{1, 4, 5, 6, 7\}, \quad f_m(t) = \min_{v \in V_m} \|tv\|,$$

and let

$$L(m) = \max_{t \in \mathbb{T}} f_m(t), \quad A(m) = \{t \in \mathbb{T} : f_m(t) = L(m)\}.$$

Our main theorem resolves the determination problem for every $m \in D$.

Theorem 1. For every $m \in D$, exactly one of the following cases applies.

1. If $11 \nmid m$, then $L(m) = 2/11$. Writing $r \equiv m \pmod{11}$,

$$A(m) = \begin{cases} \{4/11, 7/11\}, & r \in \{2, 9\}, \\ \{5/11, 6/11\}, & r \in \{3, 8\}, \\ \{4/11, 5/11, 6/11, 7/11\}, & r \in \{1, 4, 5, 6, 7, 10\}. \end{cases}$$

2. If $m = 11$, then

$$L(m) = \frac{2}{13}, \quad A(m) = \{4/13, 9/13\}.$$

3. If $m = 22$, then

$$L(m) = \frac{2}{13}, \quad A(m) = \{4/13, 6/13, 7/13, 9/13\}.$$

4. If $m = 11n$ with $n \geq 3$, then, with $Q = 11n + 4$,

$$L(m) = \frac{2n}{Q}, \quad A(m) = \left\{ \frac{5n+2}{Q}, \frac{6n+2}{Q} \right\}.$$

Consequently,

$$\{m \in D : 1/7 < L(m) < 1/6\} = \{11, 22, 33\}.$$

The proof has two ingredients. First we determine exactly the part of the graph of the base quintuple lying at or above $2/13$. This also finds every base maximizer. Second, for $m = 11n$, only two shrinking intervals around the base peaks can compete; the added runner eliminates one interval and meets the other at a single endpoint.

2 The controlling superlevel set

Define

$$g(t) = \min_{v \in \{1, 4, 5, 6, 7\}} \|vt\|.$$

The identity $g(1-t) = g(t)$ lets us work on $[0, 1/2]$.

Lemma 2 (Base-quintuple envelope). *On the half-circle,*

$$\{t \in [0, 1/2] : g(t) \geq 2/13\} = \{4/13\} \cup I_4 \cup I_5, \quad (1)$$

where

$$I_4 = \left[\frac{14}{39}, \frac{24}{65} \right], \quad I_5 = \left[\frac{41}{91}, \frac{6}{13} \right].$$

Moreover,

$$g(t) = \begin{cases} \min(6t - 2, 2 - 5t), & t \in I_4, \\ \min(7t - 3, 2 - 4t), & t \in I_5, \end{cases} \quad (2)$$

and $g(4/13) = 2/13$. Hence

$$\max_{t \in \mathbb{T}} g(t) = \frac{2}{11}, \quad \arg \max_{t \in \mathbb{T}} g(t) = \{4/11, 5/11, 6/11, 7/11\}. \quad (3)$$

Proof. The condition $\|vt\| \geq 2/13$ is equivalent to the fractional part of vt belonging to $[2/13, 11/13]$. Its allowed portions in $[0, 1/2]$ are

v	$\{t \in [0, 1/2] : \ vt\ \geq 2/13\}$
1	$[2/13, 1/2]$
4	$[1/26, 11/52] \cup [15/52, 6/13]$
5	$[2/65, 11/65] \cup [3/13, 24/65] \cup [28/65, 1/2]$
6	$[1/39, 11/78] \cup [5/26, 4/13] \cup [14/39, 37/78]$
7	$[2/91, 11/91] \cup [15/91, 24/91] \cup [4/13, 37/91] \cup [41/91, 1/2]$

Intersecting the five rows gives (1). On I_4 , the only active lower-envelope branches are $6t - 2$ and $2 - 5t$; on I_5 they are $7t - 3$ and $2 - 4t$. This proves (2), either by checking the affine branches on the indicated sawtooth cells or by direct substitution in the five circle norms.

The first triangle has its unique peak where $6t - 2 = 2 - 5t$, namely at $t = 4/11$; the second peaks where $7t - 3 = 2 - 4t$, namely at $t = 5/11$. Both peaks have height $2/11$, while the isolated point has height $2/13$. Reflection $t \mapsto 1 - t$ proves (3) and completes the equality classification. $\square$

3 Adjoining the sixth speed

We first dispose of all nonzero residue classes modulo 11.

Lemma 3. *If $11 \nmid m$, then the value and the maximizers are those in Theorem 1(1).*

Proof. By (3), $f_m(t) \leq g(t) \leq 2/11$. At the four possible equality times, the added coordinate has one value on each reflected pair:

$$\|4m/11\| \quad \text{on } \{4/11, 7/11\}, \quad \|5m/11\| \quad \text{on } \{5/11, 6/11\}.$$

For $r \equiv m \pmod{11}$, the exact residue check is

r	$11 \ 4r/11\ $	$11 \ 5r/11\ $	pairs with added norm at least $2/11$
1	4	5	both
2	3	1	$\{4/11, 7/11\}$
3	1	4	$\{5/11, 6/11\}$
4	5	2	both
5	2	3	both
6	2	3	both
7	5	2	both
8	1	4	$\{5/11, 6/11\}$
9	3	1	$\{4/11, 7/11\}$
10	4	5	both

At least one pair therefore attains $2/11$ for every nonzero residue. The base upper bound proves the value, and (3) proves that the listed points are all maximizers. $\square$

The remaining case is $m = 11n$. The next lemma treats every $n \geq 3$ at once.

Lemma 4. *Let $n \geq 3$, set $m = 11n$ and $Q = 11n + 4$. Then*

$$L(m) = \frac{2n}{Q}, \quad A(m) = \left\{ \frac{5n+2}{Q}, \frac{6n+2}{Q} \right\}.$$

Proof. Put

$$q = \frac{2n}{Q}, \quad \Delta = \frac{2}{11} - q = \frac{8}{11Q}. \quad (4)$$

For $n \geq 3$ one has $2/13 < q < 2/11$. Lemma 2 shows that $g(t) \geq q$ in $[0, 1/2]$ precisely on the two intervals

$$J_4 = \left[\frac{4}{11} - \frac{\Delta}{6}, \frac{4}{11} + \frac{\Delta}{5} \right], \quad J_5 = \left[\frac{5}{11} - \frac{\Delta}{7}, \frac{5}{11} + \frac{\Delta}{4} \right]. \quad (5)$$

If $t = 4/11 + x \in J_4$, then $11nt = 4n + 11nx$ and

$$|11nx| \leq \frac{11n\Delta}{5} = \frac{8n}{5Q} < q < \frac{1}{2}.$$

Thus $\|11nt\| = |11nx| < q$ throughout J_4 .

If $t = 5/11 + x \in J_5$, then $11nt = 5n + 11nx$, and the range of $11nx$ is

$$\left[-\frac{8n}{7Q}, \frac{2n}{Q} \right] = \left[-\frac{8n}{7Q}, q \right].$$

Both endpoint magnitudes are below $1/2$, and $8n/(7Q) < q$. It follows that $\|11nt\| \leq q$ on J_5 , with equality only at its right endpoint. At that endpoint, (2) also gives $g(t) = q$, and

$$t = \frac{5}{11} + \frac{\Delta}{4} = \frac{5n+2}{11n+4}.$$

Outside $J_4 \cup J_5$ one has $g(t) < q$. The displayed point is therefore the unique maximizer in $[0, 1/2]$. Reflection supplies the complementary time $(6n+2)/(11n+4)$ and no others. $\square$

We finally handle the two small multiples, where the isolated component in (1) survives.

Lemma 5. *For $m = 11$ and $m = 22$, the values and maximizers are those in Theorem 1(2)–(3).*

Proof. Let $n \in \{1, 2\}$, put $m = 11n$, $q_0 = 2/13$, and

$$\Delta_0 = \frac{2}{11} - q_0 = \frac{4}{143}.$$

The intervals I_4 and I_5 of Lemma 2 are equivalently

$$I_4 = \left[\frac{4}{11} - \frac{\Delta_0}{6}, \frac{4}{11} + \frac{\Delta_0}{5} \right], \quad I_5 = \left[\frac{5}{11} - \frac{\Delta_0}{7}, \frac{5}{11} + \frac{\Delta_0}{4} \right].$$

On I_4 , writing $t = 4/11 + x$, we have

$$\|11nt\| = |11nx| \leq \frac{44n}{715} < q_0.$$

On I_5 , writing $t = 5/11 + x$, the quantity $11nx$ ranges from $-44n/1001$ to $n/13$. Its magnitude is below $1/2$, and it is strictly below q_0 except at the right endpoint when $n = 2$. In that exceptional case the endpoint is $6/13$, and both the base envelope and the added norm equal q_0 .

At the isolated point in (1),

$$\|11(4/13)\| = \frac{5}{13}, \quad \|22(4/13)\| = \frac{3}{13},$$

so $t = 4/13$ attains q_0 for both values of n . Any point with $f_m(t) > q_0$ would have to lie in the interior of I_4 or I_5 , where the added norm is below q_0 ; hence $L(m) = q_0$. Equation (1) proves completeness on the half-circle, and reflection gives $\{4/13, 9/13\}$ for $n = 1$ and $\{4/13, 6/13, 7/13, 9/13\}$ for $n = 2$. $\square$

Proof of Theorem 1. Lemmas 3, 4, and 5 give the exhaustive formula and every equality time. For the final classification, $2/11 > 1/6$, whereas

$$\frac{1}{7} < \frac{2}{13} < \frac{1}{6}.$$

For $n \geq 3$, the value in Lemma 4 is greater than $2/13$ and

$$\frac{2n}{11n+4} < \frac{1}{6} \iff n < 4.$$

Thus precisely $n = 1, 2, 3$, or $m = 11, 22, 33$, satisfy both strict inequalities. $\square$

4 Exact verification and scope

The argument above is symbolic and covers every admissible parameter. As a separate reproducibility check, an exact-arithmetic Python verifier enumerates the collision/antipode candidate times justified by Cordella's finite candidate principle [1, Lemma 2.2]. Using rational arithmetic only, it recovers the three components in (1), verifies all admissible $m \leq 180$, all multipliers $n \leq 100$, and the held-out multipliers $n = 127, 251$. These finite checks support the calculation but are not used to infer the universal statement.

The use of exact phase-arc and boundary certificates has an analogue in a different operator-theoretic setting: Zeng decomposes parameter disks and phase arcs using exact rational certificates while explicitly retaining the unresolved scope of the unrestricted problem [2]. This is a methodological comparison only; that work supplies no result about lonely runners.

Computational and AI assistance. Exact rational scripts were used to check interval intersections, candidate times, values, and maximizer sets over the finite ranges stated above. OpenAI Codex assisted with proof cross-checking, literature-metadata verification, drafting, and typesetting. The author reviewed the mathematical statements and takes responsibility for the manuscript. No numerical fit, automated audit, or language-model output is used as a proof of the theorem.

Data availability. The exact-arithmetic verification script and the source used to build this article accompany the project materials. The proof itself requires no external dataset.

References

- [1] Francesco Cordella. Odd denominators in the lonely runner spectrum for six speeds, September 2026. arXiv:2609.03444v1.
- [2] Zijian Zeng. Fixed convex-lens spectral constants: Möbius reduction, sharp model theorems, and angle-dependent bounds. *Preprints*, August 2026. Version 1, posted 19 August 2026; DOI: 10.20944/preprints202608.1272.v1.